

SCQ (Single Correct Type) :

- If the equation $x^3 + bx^2 + cx + 1 = 0$ ($b < c$) has only one real root α .
Then the value of $2 \tan^{-1} (\operatorname{cosec} \alpha) + \tan^{-1} (2 \sin \alpha \sec^2 \alpha)$ is :
 (A) $-\frac{\pi}{2}$ (B) $-\pi$ (C) $\frac{\pi}{2}$ (D) π
- If semiperimeter of a triangle is 15, then the value of $(b + c) \cos(B + C) + (c + a) \cos(C + A) + (a + b) \cos(A + B)$ is equal to :
 (where symbols used have usual meanings)
 (A) -60 (B) -15
 (C) -30 (D) Can not be determined
- In triangle ABC if $A : B : C = 1 : 2 : 4$, then $(a^2 - b^2)(b^2 - c^2)(c^2 - a^2) = \lambda a^2 b^2 c^2$, where $\lambda =$
 (where notations have their usual meaning)
 (A) 1 (B) 2 (C) 4 (D) 9
- Let $f(x) = \sin x$, $g(x) = [x + 1]$ and $g(f(x)) = h(x)$,
 where $[.]$ is G.I.F. then $h' \left(\frac{\pi}{2} \right)$ is -
 (A) does not exist (B) 1 (C) -1 (D) None of these
- The number of values of x in $(0, 2\pi)$ where the function $f(x) = \frac{\tan x + \cot x}{2} - \left| \frac{\tan x - \cot x}{2} \right|$ is
 continuous but non-derivable :
 (A) 3 (B) 4 (C) 0 (D) 1
- If the two lines represented by $x + ay = b$; $z + cy = d$ and $x = a'y + b'$; $z = c'y + d'$ be perpendicular to each other, then the value of $aa' + cc'$ is :
 (A) 1 (B) 2 (C) 3 (D) 4
- Let PQ and QR be diagonals of adjacent faces of a rectangular box, with its centre at O. If $\angle QOR$, $\angle ROP$ and $\angle POQ$ are θ , ϕ and Ψ respectively then the value of ' $\cos \theta + \cos \phi + \cos \Psi$ ' is :
 (A) -2 (B) $-\sqrt{3}$ (C) -1 (D) 0
- A normal is drawn to the parabola $y^2 = 9x$ at the point $P(4, 6)$. A circle is described on SP as diameter; where S is the focus. The length of the intercept made by the circle on the normal at point P is :
 (A) $\frac{17}{4}$ (B) $\frac{15}{4}$ (C) 4 (D) 5

9. The chord of contact of a point $A(x_A, y_A)$ of $y^2 = 4x$ passes through $(3, 1)$ and point A lies on $x^2 + y^2 = 5^2$. Then:
- (A) $5x_A^2 + 24x_A + 11 = 0$ (B) $13x_A^2 + 8x_A - 21 = 0$
 (C) $5x_A^2 + 24x_A + 61 = 0$ (D) $13x_A^2 + 21x_A - 31 = 0$

MCQ (One or more than one correct) :

10. If $3 \sin \beta = \sin(2\alpha + \beta)$, then :
- (A) $(\cot \alpha + \cot(\alpha + \beta))(\cot \beta - 3 \cot(2\alpha + \beta)) = 6$
 (B) $\sin \beta = \cos(\alpha + \beta) \sin \alpha$
 (C) $\tan(\alpha + \beta) = 2 \tan \alpha$
 (D) $2 \sin \beta = \sin(\alpha + \beta) \cos \alpha$
11. Sides of a right angle triangle are $\sin x, \cos x, \tan x, x \in \left(0, \frac{\pi}{2}\right)$. Identify the correct alternative(s).
- (A) There exists an isosceles triangle satisfying above condition.
 (B) There exists a triangle satisfying above condition for which $\cos 4x = \frac{1}{4}$.
 (C) There exists a triangle satisfying above condition for which $\cos^4 x = \frac{1}{2}$.
 (D) There exists a triangle satisfying above condition whose area is $\frac{1}{4}(2^{5/4} - 2^{3/4})$.
12. Let z_1, z_2 and z_3 be three distinct complex numbers, satisfying $|z_1| = |z_2| = |z_3| = 1$. Which the following is/are true :
- (A) If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then $\arg\left(\frac{z - z_1}{z - z_2}\right) > \frac{\pi}{4}$ where $|z| > 1$
 (B) $|z_1 z_2 + z_2 z_3 + z_3 z_1| = |z_1 + z_2 + z_3|$
 (C) $\operatorname{Im}\left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 \cdot z_2 \cdot z_3}\right) = 0$
 (D) If $|z_1 - z_2| = \sqrt{2}|z_1 - z_3| = \sqrt{2}|z_2 - z_3|$, then $\operatorname{Re}\left(\frac{z_3 - z_1}{z_3 - z_2}\right) = 0$
13. If $f(x)$ is differentiable function on $\mathbb{R}$ and $g(x) = 2^{-x} f(x \cdot 2^{-x} + \lambda)$ ($\lambda \in \mathbb{R}$), then
- (A) $g'(0) + g(0) \ln 2 = f'(\lambda)$ (B) $g'(0) + g(0) \ln 2 = -f'(\lambda)$
 (C) $(2^x g(x))'(0) = -f'(\lambda)$ (D) $(2^x g(x))'(0) = f'(\lambda)$
14. If $g : \mathbb{R} \rightarrow \mathbb{N} \cup \{0\}$ and $g(x) = n$, where ' x ' represents the area of triangle joining the two fixed points $P(5, 0), Q(8, 4)$ and a variable point R such that $\angle PRQ = \frac{\pi}{2}$ and ' n ' represents the number of such triangles, then
- (A) $g(5) = 4$ (B) $g(7) = 0$ (C) $g(6.25) = 2$ (D) $g(6.25) = 1$

15. Let C be the circle $x^2 + y^2 - 6x - 8y = 0$ and P be the point (16,4). Q and R are distinct points on C such that QR is not a diameter and P,Q,R are collinear. Further L is length of chord QR and PQ + PR is an integer, if slope QR is m then
- (A) $L = 7$ (B) $RQ + PR = 25$ (C) $|m| = \frac{\sqrt{51}}{25}$ (D) $|m| = \frac{7}{24}$

Numerical Based Questions :

16. Let tangents be drawn to the curve $y^2 - 4(x + y) = 3 \sin \theta + 4 \cos \theta - 15$, $\theta \in \mathbb{R}$ from the origin when the vertex of the curve is at the maximum distance from the origin. If slopes of these tangents are m_1 and m_2 then find the value of $\left| \frac{1}{m_1 m_2} \right|$.
17. The chord AC of the parabola $y^2 = 4ax$ subtends an angle of 90° at points B and D on the parabola. If points A, B, C and D are represented by $(at_i^2, 2at_i)$, $i = 1, 2, 3, 4$ respectively, then find the value of $\left| \frac{t_2 + t_4}{t_1 + t_3} \right|$.

Paragraph Type Question :

Paragraph for Question Nos. 18 to 20

- Consider a plane $\Pi: \vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) = 5$, a line $L_1: \vec{r} = (3\hat{i} - \hat{j} + 2\hat{k}) + \lambda(2\hat{i} - 3\hat{j} - \hat{k})$ and a point A(3, -4, 1). L_2 is a line passing through A intersecting L_1 and parallel to plane Π .
18. Equation of L_2 is :
- (A) $\vec{r} = (1 + \lambda)\hat{i} + (2 - 3\lambda)\hat{j} + (1 - \lambda)\hat{k}; \lambda \in \mathbb{R}$ (B) $\vec{r} = (3 + \lambda)\hat{i} - (4 - 2\lambda)\hat{j} + (1 + 3\lambda)\hat{k}; \lambda \in \mathbb{R}$
 (C) $\vec{r} = (3 + \lambda)\hat{i} - (4 + 3\lambda)\hat{j} + (1 - \lambda)\hat{k}; \lambda \in \mathbb{R}$ (D) None of the above
19. Plane containing L_1 and L_2 is :
- (A) Parallel to yz-plane (B) Parallel to x-axis
 (C) Parallel to y-axis (D) Passing through origin
20. Line L_1 intersects plane Π at Q and xy-plane at R the volume of tetrahedron OAQR is : (where 'O' is origin)
- (A) 0 (B) $\frac{14}{3}$ (C) $\frac{3}{7}$ (D) $\frac{7}{3}$



ANSWER KEY OF CAPS – 7+

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| 1. | (B) | 2. | (C) | 3. | (A) | 4. | (A) | 5. | (B) | 6. | (A) | 7. | (C) |
| 8. | (B) | 9. | (A) | 10. | (ABCD) | 11. | (ABCD) | 12. | (BCD) | 13. | (AD) | | |
| 14. | (ABC) | 15. | (ABC) | 16. | 4 | 17. | 1 | 18. | (C) | 19. | (B) | 20. | (D) |